

Node Structure Visualization

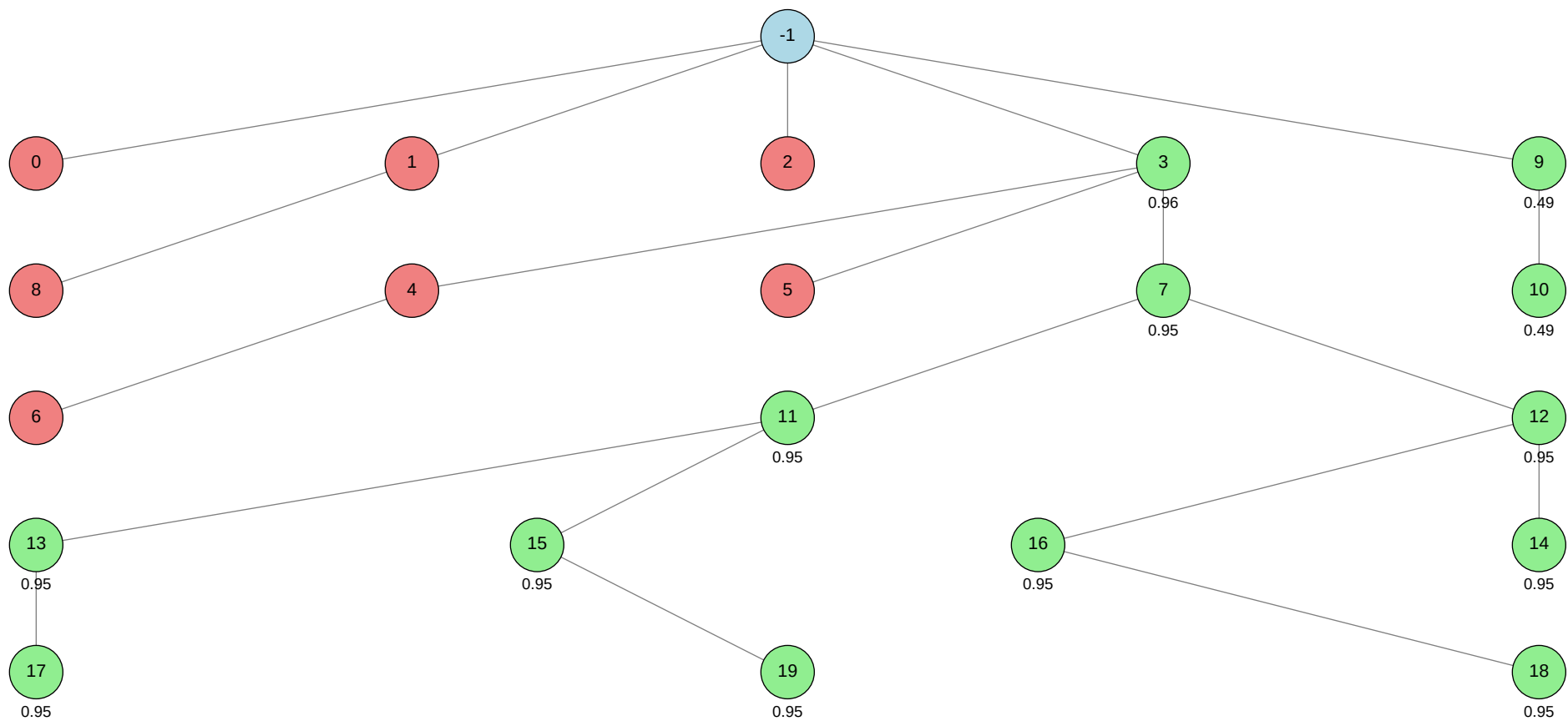
Click on any node in the tree to jump to its detailed information.

Total Nodes: 21 | Best Validation Score: 0.9606 (Node 3)

Node Tree:

Table of Contents:

Node ID	Stage	Status	Tool
<-1			
>0			autogluon.multimodal
>1			machine learning
>2			autogluon.tabular
>3			autogluon.multimodal
>4			autogluon.multimodal
>5			autogluon.multimodal
>6			autogluon.multimodal
>7			autogluon.multimodal
>8			machine learning
>9			machine learning
>10			machine learning
>11			autogluon.multimodal
>12			autogluon.multimodal
>13			autogluon.multimodal
>14			autogluon.multimodal
>15			autogluon.multimodal
>16			autogluon.multimodal
>17			autogluon.multimodal
>18			autogluon.multimodal
>19			autogluon.multimodal



Status	Color
Success	Green
Failure	Red
Neutral	Blue

Node Details:

Node -1 (root)

Property	Value
uct_value	0.4189
ctime	2026-05-22 16:55:16
debug_attempts	0
depth	0
execution_time	0.0
expected_child_count	0
failure_visits	7
id	-1
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✗
num_children	5
prev_tutorial_prompt	None
stage	root
time_step	-1
tool_used	
tools_available	[]
unvalidated_visits	0
validated_reward	11.490600000000002
validated_visits	13
validation_score	None
visits	20
parent_id	None
child_ids	[0, 1, 2, 3, 9]

Node 0 (evolve)

Property	Value
uct_value	2.4474
ctime	2026-05-22 16:56:14
debug_attempts	0
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	1
id	0
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	None
stage	evolve
time_step	0
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	1
parent_id	-1
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 34-basic_prompt/node_0/conda_env

Resolved 235 packages in 5.22s

Downloading pydantic-core (2.0MiB)

Downloaded pydantic-core

Prepared 10 packages in 2.81s

Uninstalled 1 package in 2.05s

Installed 233 packages in 1m 11s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10

+ catboost==1.2.10
+ certifi==2026.5.20
+ cffi==2.0.0
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ chronos-forecasting==2.2.2
+ click==8.4.1
+ cloudpathlib==0.24.0
+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ ... [truncated, 17248 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the test set (`test_pred_df`) does not contain the columns `skin_tone`, `alternative_skin_tone`, and `expert_opinion`, which were present in the training set and included as features (`feature_cols`). When the code tries to select these columns from the test DataFrame for prediction, a `KeyError` is raised since they are missing from the test data.

SUGGESTED_FIX:

Update the code to only use features that are present in both the training and test sets when making predictions. Specifically, before selecting `feature_cols` from the test DataFrame, intersect `feature_cols` with the columns available in `test_pred_df` to avoid referencing missing columns; ensure that the prediction input matches the test set's actual structure.

Node 1 (evolve)

Property	Value
uct_value	1.7306
ctime	2026-05-22 17:42:37
debug_attempts	1
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.1115599719432814
failure_penalty	-0.0
failure_visits	2
id	1
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	None
stage	evolve
time_step	1
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	2
parent_id	-1
child_ids	[8]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

warning: Requirements file `~/home/anri21/.conda/envs/mlzero\_env/lib/python3.11/site-packages/autogluon/assistant/tools\_registry/machine learning/requirements.txt` does not contain any dependencies

Using Python 3.11.15 environment at: 34-basic_prompt/node_1/conda_env

Resolved 79 packages in 625ms

Uninstalled 1 package in 1.73s

Installed 78 packages in 36.59s

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiosignal==1.4.0

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ anyio==4.13.0

+ attrs==26.1.0

+ certifi==2026.5.20

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ click==8.4.1

+ cuda-bindings==13.2.0

+ cuda-pathfinder==1.5.4

+ cuda-toolkit==13.0.2

+ datasets==4.8.5

+ dill==0.4.1

+ filelock==3.29.0

+ frozenlist==1.8.0

+ fsspec==2026.2.0

+ h11==0.16.0

+ hf-xet==1.5.0

+ httpcore==1.0.9

+ httpx==0.28.1

+ huggingface-hub==1.16.1

+ idna==3.16

+ jinja2==3.1.6

```
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.14.0a1
+ pydantic-core==2.47.0
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+... [truncated, 2888 chars total]
```

Error Analysis:

ERROR_SUMMARY: The error occurs because the version of LightGBM installed in your environment does not support passing `early_stopping_rounds` directly as a keyword argument to `lgb.train()`; this argument is only supported in newer versions or via the scikit-learn API (`LGBMClassifier.fit`), or should be implemented using callbacks in older versions.

SUGGESTED_FIX: Remove the `early_stopping_rounds` argument from the `lgb.train()` call and instead use the `callbacks` parameter with `lgb.early_stopping(50)`, or upgrade LightGBM to a version that supports `early_stopping_rounds` directly. Alternatively, switch to using `LGBMClassifier` from `lightgbm.sklearn`, which supports `early_stopping_rounds` in its `fit()` method.

Node 2 (evolve)

Property	Value
uct_value	2.4474
ctime	2026-05-22 17:47:01
debug_attempts	0
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2131672923786587
failure_penalty	-0.0
failure_visits	1
id	2
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	None
stage	evolve
time_step	2
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	1
parent_id	-1
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 34-basic_prompt/node_2/conda_env

Resolved 239 packages in 1.18s

Uninstalled 1 package in 1.62s

Installed 237 packages in 49.25s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.1
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + daal==2025.9.0
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0
- + httpx==1.0.dev3
- + huggingface-hub==0.36.2

+ hyperopt==... [truncated, 41767 chars total]

Error Analysis:

ERROR_SUMMARY: AutoGluon failed to train any models because the Ray backend (used for parallel model training) attempted to create UNIX domain sockets with file paths exceeding the system limit of 107 bytes. This resulted in repeated OSError exceptions ("validate_socket_filename failed: AF_UNIX path length cannot exceed 107 bytes"), causing all model training to be skipped and ultimately raising a RuntimeError due to no models being fit.

SUGGESTED_FIX:

Shorten the directory paths used for temporary files and model outputs, especially the base working directory and any subfolders under /sc-scratch/sc-scratch-ikim-guidlight/. Move your project to a location with a much shorter absolute path (e.g., /tmp/mlzero/ or /scratch/mlzero/) and update all relevant script and data paths accordingly. This will ensure that Ray's socket file paths remain under the 107-byte limit and allow model training to proceed.

Node 3 (debug)

Property	Value
uct_value	1.3954
avg_raw_score	0.95547
ctime	2026-05-22 17:52:58
debug_attempts	1
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.7223749447744343
exploration	0.6729441351364719
failure_penalty	-0.038461538461538464
failure_visits	3
id	3
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	1
normalized_score	0.9890874282067645
num_children	3
prev_tutorial_prompt	None
stage	debug
time_step	3
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.7608364832359727
validated_reward	10.5096

validated_visits	11
validated_weight	0.7692307692307693
validation_score	0.9606
visits	14
parent_id	-1
child_ids	[4, 5, 7]

Node 4 (evolve)

Property	Value
uct_value	1.6243
ctime	2026-05-22 18:14:34
debug_attempts	1
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.6013040756694474
failure_penalty	-0.0
failure_visits	2
id	4
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✗
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	4
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0

validated_visits	0
validation_score	None
visits	2
parent_id	3
child_ids	[6]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 34-basic_prompt/node_4/conda_env

Resolved 235 packages in 1.23s

Uninstalled 1 package in 1.88s

Installed 233 packages in 1m 15s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10

- + catboost==1.2.10
- + certifi==2026.5.20
- + cffi==2.0.0
- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.1
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0

+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.1... [truncated, 12100 chars total]

Error Analysis:

ERROR_SUMMARY: The script failed with a KeyError because the hyperparameter 'env.num_cpus' was included in the hyperparameters dictionary, but this key does not exist in the AutoGluon MultiModalPredictor configuration. Additionally, several hyperparameters use deprecated names with the 'optimization.' prefix instead of the updated 'optim.' prefix, which leads to warnings and may cause further compatibility issues.

SUGGESTED_FIX: Remove 'env.num_cpus' from the hyperparameters dictionary, as it is not a valid option for MultiModalPredictor. Update all hyperparameter keys that use the 'optimization.' prefix to use the correct 'optim.' prefix (e.g., change 'optimization.learning_rate' to 'optim.lr', 'optimization.lr_scheduler' to 'optim.lr_scheduler', etc.). Double-check the AutoGluon documentation for the correct hyperparameter names and ensure that only supported keys are included.

Node 5 (evolve)

Property	Value
uct_value	2.2971
ctime	2026-05-22 18:22:24
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.664857771836881
failure_penalty	-0.0
failure_visits	1
id	5
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	5
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0

validated_visits	0
validation_score	None
visits	1
parent_id	3
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 34-basic_prompt/node_5/conda_env

Resolved 235 packages in 1.24s

Uninstalled 1 package in 1.83s

Installed 233 packages in 1m 15s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10

- + catboost==1.2.10
- + certifi==2026.5.20
- + cffi==2.0.0
- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.1
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0

+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.1... [truncated, 9994 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the hyperparameter "optim.lr_scheduler" is not found in the AutoGluon MultiModal configuration. This means that the key "optim.lr_scheduler" (with value "cosine") is not recognized or supported by the current version of AutoGluon you are using, causing a KeyError during model training.

SUGGESTED_FIX: Remove the "optim.lr_scheduler" entry from your hyperparameters dictionary and try running the script again. If you wish to control the learning rate schedule, use the supported key "optim.lr_schedule" (e.g., "cosine_decay", "polynomial_decay", or "linear_decay") as per the official AutoGluon documentation and tutorials.

Node 6 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-22 18:30:13
debug_attempts	1
depth	4
execution_time	0.0
expected_child_count	0
failure_visits	1
id	6
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	6
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	4
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 34-basic_prompt/node_6/conda_env

Resolved 235 packages in 1.26s

Uninstalled 1 package in 1.90s

Installed 233 packages in 57.32s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ chronos-forecasting==2.2.2

+ click==8.4.1

+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.1... [truncated, 10171 chars total]

Error Analysis:

ERROR_SUMMARY: The script failed with a `KeyError` because the hyperparameter `"optim.monitor"` (and related keys like `"optim.metric\_for\_best\_model"` and `"optim.save\_best"`) does not exist in the current AutoGluon MultiModal configuration. These keys were included in the `hyperparameters` dictionary, but AutoGluon

does not recognize them, resulting in an error during model training.

SUGGESTED_FIX:

Remove the keys `"optim.monitor"`, `"optim.metric_for_best_model"`, and `"optim.save_best"` from the `hyperparameters` dictionary in your script. Only include hyperparameters that are documented and supported by your installed version of AutoGluon MultiModal; refer to the official documentation or use only those keys shown in the provided tutorials (such as learning rate, epochs, batch size, etc.). After removing these unsupported keys, rerun your script.

Node 7 (debug)

Property	Value
uct_value	1.7143
avg_raw_score	0.9549000000000001
ctime	2026-05-22 18:38:32
debug_attempts	1
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.9878749202297386
exploration	0.7548619804316817
failure_penalty	-0.0
failure_visits	0
id	7
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9878749202297386
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	7
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9878749202297386
validated_reward	9.5490000000000001
validated_visits	10
validated_weight	1.0
validation_score	0.9549
visits	10
parent_id	3
child_ids	[11, 12]

Node 8 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-22 18:56:31
debug_attempts	1
depth	2
execution_time	0.0
expected_child_count	0
failure_visits	1
id	8
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	<pre>### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have the necessary tools and knowledge to complete the tasks. # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have the necessary tools and knowledge to complete the tasks.</pre>
stage	debug
time_step	8
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0

validation_score	None
visits	1
parent_id	1
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

warning: Requirements file `~/home/anri21/.conda/envs/mlzero\_env/lib/python3.11/site-packages/autogluon/assistant/tools\_registry/machine learning/requirements.txt` does not contain any dependencies

Using Python 3.11.15 environment at: 34-basic_prompt/node_8/conda_env

Resolved 79 packages in 1.17s

Uninstalled 1 package in 1.67s

Installed 78 packages in 46.80s

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiosignal==1.4.0

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ anyio==4.13.0

+ attrs==26.1.0

+ certifi==2026.5.20

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ click==8.4.1

+ cuda-bindings==13.2.0

+ cuda-pathfinder==1.5.4

+ cuda-toolkit==13.0.2

+ datasets==4.8.5

+ dill==0.4.1

+ filelock==3.29.0

+ frozenlist==1.8.0

+ fsspec==2026.2.0

+ h11==0.16.0

+ hf-xet==1.5.0

+ httpcore==1.0.9

+ httpx==0.28.1

+ huggingface-hub==1.16.1

+ idna==3.16

```
+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.14.0a1
+ pydantic-core==2.47.0
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+... [truncated, 2814 chars total]
```

Error Analysis:

ERROR_SUMMARY: The error occurs because the `lgb.train()` function in LightGBM no longer accepts the `verbose_eval` argument when using the callbacks API (such as `lgb.early_stopping`). Passing `verbose_eval` as a keyword argument causes a `TypeError` in recent LightGBM versions.

SUGGESTED_FIX: Remove the `verbose_eval` argument from the `lgb.train()` call. If you want to control logging frequency, use the `lgb.log_evaluation()` callback instead by adding it to the callbacks list.

Node 9 (debug)

Property	Value
uct_value	1.7306
avg_raw_score	0.4905
ctime	2026-05-22 19:01:07
debug_attempts	2
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2867582287320087
failure_penalty	-0.0
failure_visits	0
id	9
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.0
num_children	1
prev_tutorial_prompt	None
stage	debug
time_step	9
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.981

validated_visits	2
validated_weight	1.0
validation_score	0.4905
visits	2
parent_id	-1
child_ids	[10]

Node 10 (evolve)

Property	Value
uct_value	1.1772
ctime	2026-05-22 19:05:54
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	0
id	10
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	<pre>### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them</pre>
stage	evolve
time_step	10
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.4905
validated_visits	1

validation_score	0.4905
visits	1
parent_id	9
child_ids	[]

Node 11 (evolve)

Property	Value
uct_value	1.9474
avg_raw_score	0.9549
ctime	2026-05-22 19:10:40
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
exploitation	0.9878749202297382
exploration	1.0479887918088304
failure_penalty	-0.0
failure_visits	0
id	11
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9878749202297382
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	11
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9878749202297382
validated_reward	4.7745
validated_visits	5
validated_weight	1.0
validation_score	0.9549
visits	5
parent_id	7
child_ids	[13, 15]

Node 12 (evolve)

Property	Value
uct_value	2.0607
avg_raw_score	0.9549
ctime	2026-05-22 19:27:14
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
exploitation	0.9878749202297382
exploration	1.0479887918088304
failure_penalty	-0.0
failure_visits	0
id	12
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9878749202297382
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	12
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9878749202297382
validated_reward	3.8196
validated_visits	4
validated_weight	1.0
validation_score	0.9549
visits	4
parent_id	7
child_ids	[16, 14]

Node 13 (evolve)

Property	Value
uct_value	2.2563
avg_raw_score	0.9549
ctime	2026-05-22 19:43:06
debug_attempts	0
depth	6
execution_time	0.0
expected_child_count	0
exploitation	0.9878749202297382
exploration	1.1772322201769845
failure_penalty	-0.0
failure_visits	0
id	13
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9878749202297382
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	13
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9878749202297382
validated_reward	1.9098
validated_visits	2
validated_weight	1.0
validation_score	0.9549
visits	2
parent_id	11
child_ids	[17]

Node 14 (evolve)

Property	Value
uct_value	2.6527
avg_raw_score	0.9549
ctime	2026-05-22 19:58:49
debug_attempts	0
depth	6
execution_time	0.0
expected_child_count	0
exploitation	0.9878749202297382
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	0
id	14
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9878749202297382
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	14
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9878749202297382
validated_reward	0.9549
validated_visits	1
validated_weight	1.0
validation_score	0.9549
visits	1
parent_id	12
child_ids	[]

Node 15 (evolve)

Property	Value
uct_value	2.2563
avg_raw_score	0.9549
ctime	2026-05-22 20:14:38
debug_attempts	0
depth	6
execution_time	0.0
expected_child_count	0
exploitation	0.9878749202297382
exploration	1.664857771836881
failure_penalty	-0.0
failure_visits	0
id	15
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9878749202297382
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	15
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9878749202297382
validated_reward	1.9098
validated_visits	2
validated_weight	1.0
validation_score	0.9549
visits	2
parent_id	11
child_ids	[19]

Node 16 (evolve)

Property	Value
uct_value	2.1651
avg_raw_score	0.9549
ctime	2026-05-22 20:30:59
debug_attempts	0
depth	6
execution_time	0.0
expected_child_count	0
exploitation	0.9878749202297382
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	0
id	16
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9878749202297382
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	16
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9878749202297382
validated_reward	1.9098
validated_visits	2
validated_weight	1.0
validation_score	0.9549
visits	2
parent_id	12
child_ids	[18]

Node 17 (evolve)

Property	Value
uct_value	2.1651
ctime	2026-05-22 20:47:01
debug_attempts	0
depth	7
execution_time	0.0
expected_child_count	0
failure_visits	0
id	17
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	17
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.9549
validated_visits	1
validation_score	0.9549
visits	1
parent_id	13
child_ids	[]

Node 18 (evolve)

Property	Value
uct_value	2.1651
ctime	2026-05-22 21:03:55
debug_attempts	0
depth	7
execution_time	0.0
expected_child_count	0
failure_visits	0
id	18
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	18
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.9549
validated_visits	1
validation_score	0.9549
visits	1
parent_id	16
child_ids	[]

Node 19 (evolve)

Property	Value
uct_value	2.1651
ctime	2026-05-22 21:19:53
debug_attempts	0
depth	7
execution_time	0.0
expected_child_count	0
failure_visits	0
id	19
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	19
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.9549
validated_visits	1
validation_score	0.9549
visits	1
parent_id	15
child_ids	[]

